

— AGENTIC PDLC —

Agentic PDLc.

Raising the **quality bar** and **accelerating delivery** for MedTech.

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The Project Console

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Submission Package · Gap Analysis ·
Value & ROI · Project Settings ·
Workflows

01

Agentic PDLC Overview

What the operating model is, what it changes, and why the regulated stack benefits first.

STAT DASHBOARD

Three external signals. Three different domains.



PRODUCTIVITY FLOOR

~55% faster

Copilot autocomplete on coding tasks.

The narrowest measurable case — autocomplete in code. Broader agentic delivery operates on a much larger work surface.[1] GitHub Research, 2022



ENTERPRISE STUDIES CONVERGE

10–40% uplift

Task-level gains across coding, content, analysis.

Annual productivity uplift in tech & engineering at the high end across function categories.
[2] McKinsey • [5] Stanford AI Index



REGULATORS AHEAD OF INDUSTRY

FDA PCCP

Final guidance • December 2024.

AI-enabled devices may be updated under a pre-specified change-control plan — without re-submitting on every retrain.[3][4] FDA • 2021/2024

COST OF WAITING

An organization that delays adoption will be 18–24 months behind peers that invest now. The asymmetry compounds — every month widens the gap.

OUR DIFFERENCE

Most “agentic” MedTech is a wrapper. Ours is a purpose-built harness.

WHAT VENDORS SHIP

Context and a UI wrapped around someone else’s model.

1 • Code-completion rebranded

Copilot, Cursor, Codeium with a vendor wrapper. **Productive on the keystroke — blind to everything else.** No idea your work product needs IEC 62304 traceability, that an audit trail must name a human, or that an output is going into a 510(k).

2 • Chatbot on a data lake

RAG over your docs. **No provenance from answer to source** — you can’t tell which document fragment shaped the response. Embeddings shift on every rebuild; data ages silently. *Same question, different answer next month.* No full-context grounding — just whatever the nearest-neighbor lookup returns.

3 • Hosted autonomous-agent studio

A closed-box agent running in someone else’s cloud. **The audit trail is theirs, not yours. Not purpose-built for MedTech** — no IEC 62304, ISO 14971, or DHF grounding. **Not portable** — change vendors and both the capability and your history walk away.

WHAT WE BUILT

A purpose-built MedTech agentic harness — versioned, audited, and yours.

- ▶ **Domain-grounded.** IEC 62304, ISO 14971, and ISO 13485 encoded directly into skills, agents, hooks, and rules — with a Part 11-aware publishing flow into the formal review system.
- ▶ **Audit-trailed by construction.** Every decision attributable; every change links design inputs to verification to the DHF record.
- ▶ **Lives in your repository.** Versioned alongside the device. No vendor tenant. No platform you have to log into.
- ▶ **Built for regulated delivery.** DHF-shaped outputs, submission-ready by construction, reusable across programs in your portfolio.

THE CAPABILITY MATRIX

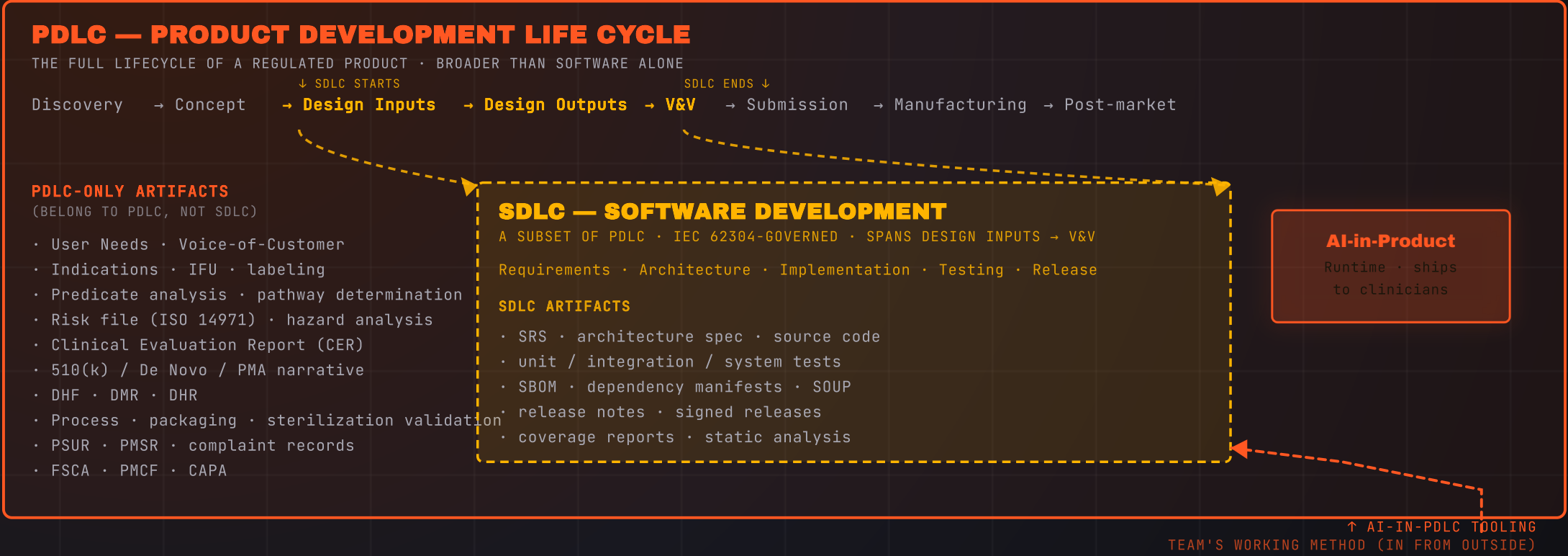
Six capabilities. One bucket has them all.

	01 CODE-COMPLETION REBRANDED	02 CHATBOT ON A DATA LAKE	03 HOSTED AUTONOMOUS- AGENT STUDIO	04 PURPOSE-BUILT MEDTECH HARNESS
Productivity gains across coding, documentation, and analysis	✓	✓	✓	✓
Process / workflow enforcement	✗	✗	~	✓
Domain grounding (regulatory · clinical)	✗	✗	~	✓
Audit trail in <i>your</i> hands	✗	✗	✗	✓
Lives in your repository · versioned	✗	✗	✗	✓
Cross-program IP propagation (registry)	✗	✗	~	✓

✓ present · ~ partial / vendor-mediated · ✗ structurally absent

NESTED CONTAINERS

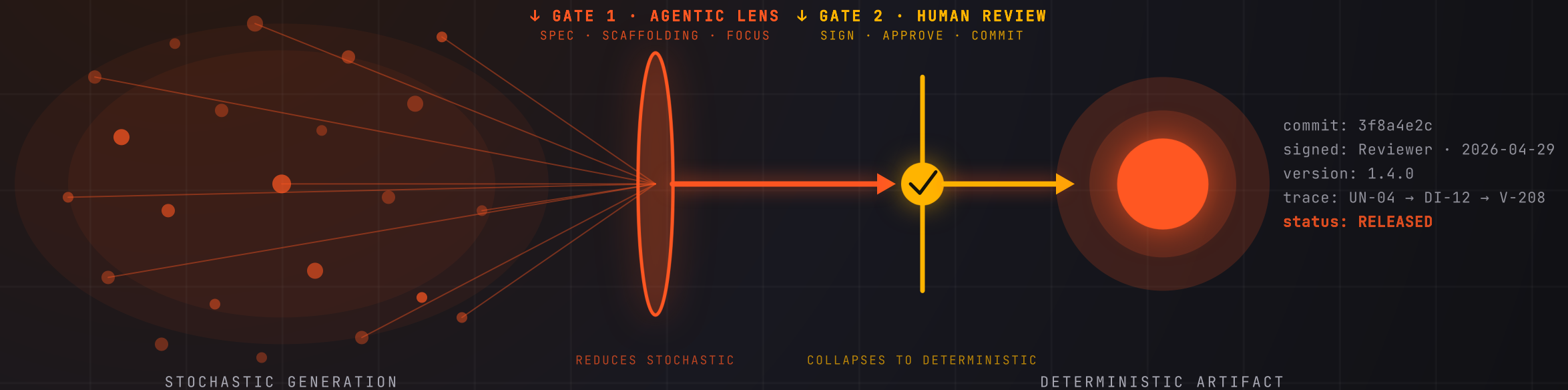
SDLC sits *inside* PDLC. AI-in-Product is a runtime resident.



Containment, not equivalence. Most "AI in delivery" pitches conflate these three. The agentic PDLC harness operates on the team's working method — the dashed arrow that enters the lifecycle from outside.

THE WAVE COLLAPSES AT COMMIT

Stochastic at generation. Deterministic at commit.



The probabilistic step is an authoring step that happens once per change. By the time a regulator sees the artifact, there is no probability left in it. *A regulator is not approving the model — a regulator is approving the artifact.*

02

What is Agent Engineering?

Above autocomplete-with-chat. The conductor, the orchestra, the score, and the six layers of guardrails.

THE LOAD-BEARING ANALOGY

A bulb gives you light. Optics give you a laser.



The GlobalLogic Agentic PDLC harness is the optics — not the bulb. Anyone can rent a bulb.

01

Skills

Packaged, reusable expert checklists. Same shape, every time.

02

Rules

Short conventions the assistant reads and obeys.

03

Agents

Domain personas with citation discipline.

04

Hooks

Automations on session start, before edits, on close.

05

Registries

Versioned catalogs. Cross-project propagation.

06

Guardrails

Layered defense — pre-action, in-flight, post-action, release.

MENTAL MODEL #1 — ORGANIZATIONAL

The Conductor — what the human does.



01 · SCORE

Sheet music



Specs. Skills · rules · rubrics. The score travels under any orchestra.

02 · CONDUCTOR

The lead



Human in the loop. Calibration Mode + Domain Mode. Same person, two distinct acts.

03 · ORCHESTRA

Musicians



Model + advisors. Capable, fast, can play many parts in parallel. Doesn't know the piece.

04 · PERFORMANCE

The concert



The artifact the customer receives. Code · DHF · trace matrix · submission narrative.

05 · RECORDING

The tape



Audit trail. Sync · lessons · digest. Feeds back into the score.

Conductor's two modes: **DOMAIN MODE** (knows the piece — outside regulatory + clinical knowledge) · **CALIBRATION MODE** (hears drift — codifies corrections into specs).

THE INVESTMENT RANKING

Specs first. Scaffolding second. Model third.

01 Specs

LARGEST VALUE DELTA

Every repeatable quality improvement in the corpus was a spec improvement. Every recurring complaint becomes a versioned spec change *before* any model-tuning is attempted.

02 Scaffolding

MEDIUM VALUE DELTA · INFRASTRUCTURE

Skills · hooks · registries · idempotency contracts. The infrastructure that makes specs operationally meaningful and propagatable across program work.

03 Model choice

TUNING LEVER ONLY

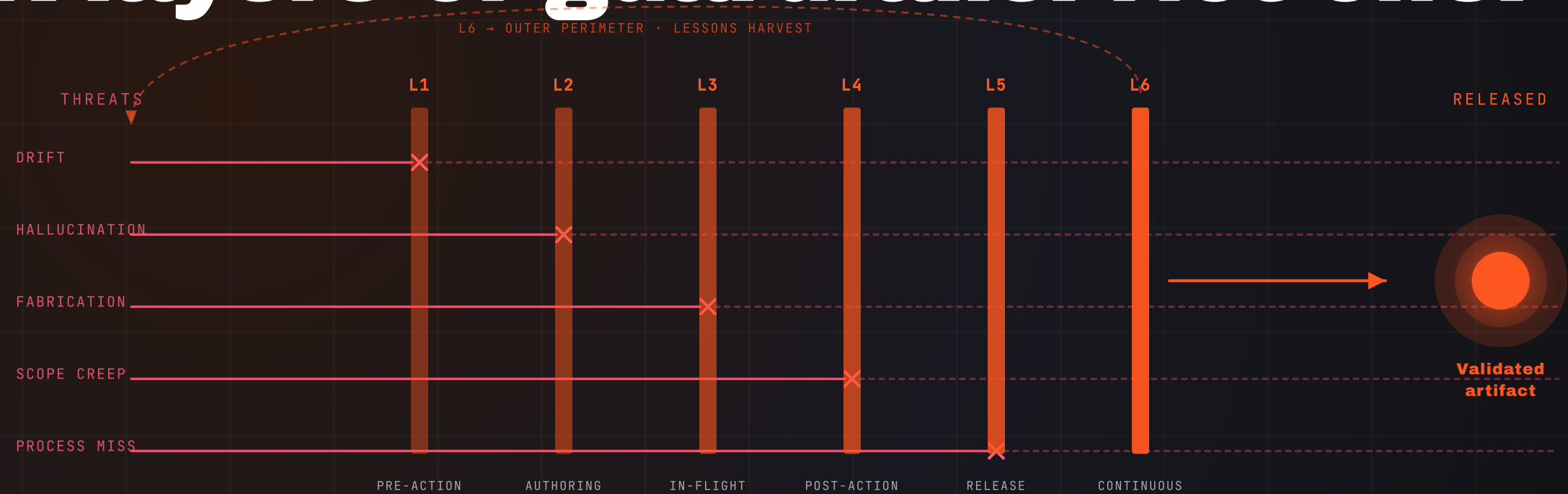
Real, but smallest. Bigger model for harder synthesis; faster model for high-volume routine work.

THE INVARIANT

A score performs the same way under any competent orchestra. A spec produces the same artifact under any competent model. *The score travels.*

THE DEFENSE-IN-DEPTH

Six layers of guardrails. Not one.



In an unregulated context, a single L1 gate is often enough. In HCLS, any single layer's failure means the entire compliance posture collapses. Multi-model substrate is part of the layered defense — adversarial validation requires *different* models checking each other.

- L1

Pre-action gates

Block before the action.
Task gate · freeze gate.
- L2

Authoring discipline

CLAUDE.md rules · [VERIFY]
· default-to-action.
- L3

In-flight verification

Advisor counterpoint ·
multi-perspective panels.
- L4

Post-action checks

Best-practices audit ·
regression-guard lint.
- L5

Release gates

Named-reviewer
attestation · Comala (Part
11).
- L6

Continuous improvement

Lessons harvest · digest ·
sync-log.

03

Sample Project Overview & Strategy

PDLC_DEMO · PainEase PCA Advanced — the device, the regulatory strategy, and the deliverables shape.

STRATEGY HARNESS · IN ACTION

Strategies are managed artifacts — the basis for everything downstream.

Regulatory, clinical, architectural, post-market — each strategy is a curated, versioned artifact in the harness, the canonical reference every downstream document derives from.

↓ What follows are generated outputs from the sample project — a working strategy harness in action.

What PP3500 is

PP3500 — **PainEase PCA Advanced** — is a patient-controlled analgesia infusion pump. The program as modeled in this repo is a **single composite system** of SaMD + SiMD + custom medical-electrical hardware, with adjacent infrastructure around it:

DHF (ARCHITECTURE NAME)	CLASSIFICATION	COMPOSITION	WHAT IT DOES
PCA Device (pca-device)	Class II medical device	SiMD pump firmware + SaMD modules + hardware	On-pump delivery, dose-limit enforcement, alarms, UI, self-test
Connectivity Adapter (connectivity-adapter)	MDDS (non-device post-2015 FDA)	SaMD	On-prem bridge that carries PCA telemetry + drug-library pushes to/from the Cloud Suite
Cloud Suite (cloud-suite)	Mixed — see children	SaMD + non-device software	Platform DHF. Seven child DHFs carry the regulatory weight individually

Regulatory strategy at a glance

We are filing **one 510(k)** for the PCA device itself, **with an integrated PCCP** that pre-authorizes two classes of post-market change: drug-library updates and bounded firmware updates.

Filing scope = PCA device alone.

The Connectivity Adapter (MDDS) and all Cloud Suite components except the Drug Library Manager are **out of the PP3500 filing**. Each is analyzed and filed (or not filed) on its own regulatory merits. See the regulatory strategy, §1 *Filing Scope: PCA Device Alone*.

Critical-requirement carve-out (CtS / CtF / CtC / CtP).

Every user need and design input is tagged with one or more of **Critical to Safety / Function / Compliance / Performance**. Only Ct*-tagged requirements are in scope for the 510(k) + PCCP; everything else ships post-clearance on the commercial track. This keeps the submission minimum and the V&V burden tractable. See the regulatory strategy, §1 *Filing Strategy — Critical-Requirement Carve-out*.

PCCP envelope.

The PCCP pre-authorizes changes to Ct*-tagged requirements that stay within pre-specified bounds — drug-library updates, firmware patches against a fixed risk profile, and predictive-alarm SaMD updates that meet the PCCP change-protocol acceptance criteria.

Predicate.

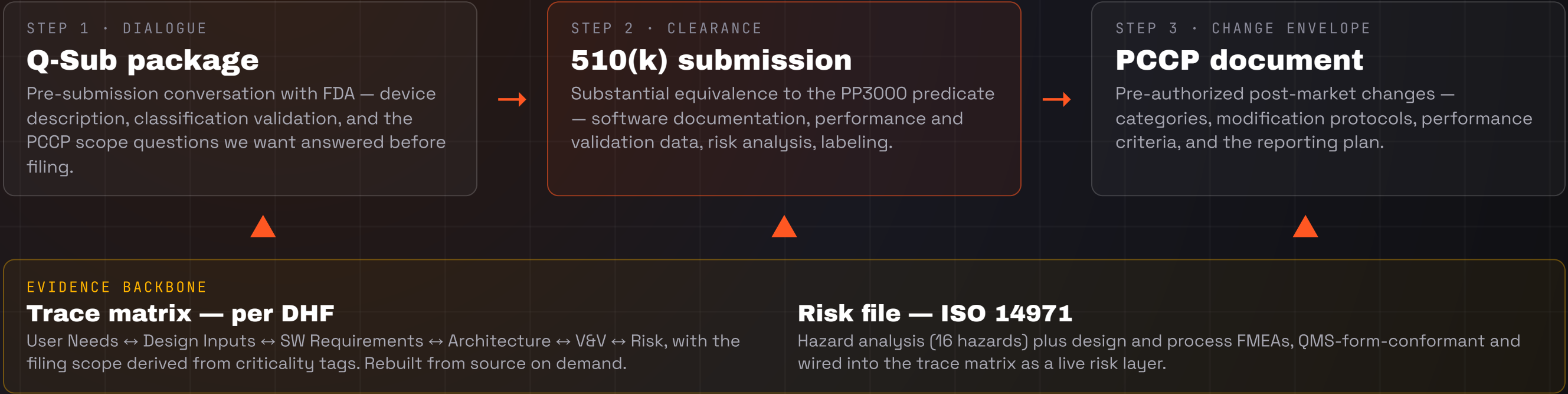
PP3000 (K190567) — a prior PainEase PCA pump in the same family. The PCCP envelope is the substantive delta vs. the predicate.

Drug Library Manager

classified as Class II SaMD accessory, is either bundled into the PP3500 filing or filed separately (posture TBD in submission planning — see the 510(k) composition manifest).

KEY DELIVERABLES

Three filings. One evidence backbone.



Every filing draws on the **same** trace and risk evidence — built once, cited everywhere, never copy-pasted.

04

Project Shape & Capabilities

Operating rules, project-level skills, and persona agents — the machinery underneath the work.

OPERATING RULES · BAKED INTO THE PROJECT

Nine rules the tooling enforces.

Task-first gate

No file edit without an owning task document — a hook refuses otherwise. No orphan changes, ever.

Session security check

Every session opens with an audit against the project manifest’s tool allowlists; drift triggers remediation.

Local semantic search

Ranked search over the whole documentation corpus, fully local — no cloud calls, one shared index.

One task, one file

All analysis, drafts, and decisions live inside the task’s own document. The trail stays auditable.

Checkpoint & recovery

A session that ends without a checkpoint leaves a marker; the next session recovers the narrative from git history.

Usage telemetry

Token spend is collected per session and published through git into the console’s metrics view.

Real-time capture

Strategy decisions and lessons land in the task doc the same turn they happen — not in a cleanup pass.

Audited conversions

Raw document-conversion commands are blocked — every DOCX/PDF round-trip goes through one audited pipeline.

Single source of truth

Identity, DHF topology, team roster, registries, and security policy live in one project manifest.

SKILLS · PACKAGED EXPERT PLAYBOOKS

35 skills. These carry the program.

Every skill is allow-listed in the project manifest and pulled from an auditable registry — the same playbook, run the same way, by everyone.

AUTHOR

medtech-docs

Scaffolds the DHF and documentation tree per FDA / IEC 62304 expectations.

docflow

Audited DOCX / PDF / XLSX round-trips — images, cross-references, metadata preserved.

submissions

Builds the Q-Sub / 510(k) / PCCP package structure and its composition manifests.

regulatory-authoring

Lint, copy-edit, and QA-conformance for every regulator-facing sentence.

VERIFY

trace-matrix

Six-layer bidirectional trace, rebuilt from source on demand.

dhf-manifest

Are the right documents present for the regulatory obligations we carry?

gap-analysis

Critiques our own work product against the standards it claims to meet.

red-team

A hostile buyer committee stress-tests outward documents before they ship.

reference-audit

Every citation independently re-derived from the byte-correct source.

OPERATE

task

The task-first gate: no change without an owning task document.

tracker

Milestone-driven submission-readiness dashboard, editable in the browser.

project-console

The browser workbench over the same agents and artifacts.

usage-metrics

Token cost and modeled hours-saved, per task, honestly framed.

lessons

Corrections harvested, staged, and promoted into permanent guardrails.

AGENTS · THE ADVISORY BENCH

One bench. Thirty-one advisors. Four jobs.

CORE TEAM — 11 ASSISTANTS

regulatory-affairs · clinical-affairs · risk-management ...

One specialist per discipline — regulatory, clinical, risk, cybersecurity, quality, V&V, human factors, systems, R&D, post-market, program — each grounded in the project’s own DHF and strategies, citing its sources.

KOL PERSONAS — 8 VOICES

clinicians & domain experts

Outside clinical voices that pressure-test user needs, workflows, and claims the way a real advisory board would.

RED TEAM — 7 SKEPTICS

CEO · CFO · CTO · VP-Eng · QA-VP · RA-VP · PMO

A hostile buyer committee. Every outward-facing document faces them — and their objections — before a real buyer ever sees it.

PANELS — 5 BOARDS

core-team panel · design-review panel ...

Cross-functional boards that answer as one — program questions get multi-perspective input in a single pass.

Same agents, two runtimes. One definition serves both the engineering environment and the browser console — and a security agent audits every session against the project manifest. Updating an advisor updates every surface at once.

05

Quality & Process

How the guardrails actually work — hooks, traces, and the audit surface a regulator can read.

SKILLS · THE FULL TOOLBOX

Thirty-five playbooks. One auditable registry.

DHF & REGULATORY	QUALITY & REVIEW	PROGRAM OPS	AUTHORING & FORMATS	PLATFORM
medtech-docs	gap-analysis	task	docx	file-locator
docflow	red-team	tracker	pptx	skill-creator
dhf-manifest	best-practices	strategy	xlsx	sync-skills
trace-matrix	secops	digest	pdf	web-control
submissions	writing-well	jira-pull	md-deck	
change-control	lessons	usage-metrics	frontend-design	
regulatory-authoring		project-console	frontend-slides	
reference-audit		advisors	explain	
			knowledge-pack-export	

Every skill is **allow-listed** in the project manifest and synced from a shared registry — consistency is installed, not trained.

Hooks — automatic tripwires you can't forget to run

A **hook** is a small automation that fires at a specific moment — when a session starts, when a file is about to be edited, when work is about to end — and either allows, blocks, or adds context to the action. Hooks are invisible until one trips. When one trips, it means the system caught an issue before it became a problem.

HOOK	WHEN IT FIRES	WHAT IT PREVENTS
Task-first gate (check-active-task.sh)	Before any file edit	Refuses to edit a file unless there's an active task document that owns the change. No orphan edits, ever.
Session security check (security-assert.sh)	When a session opens	Runs the project-secops audit against project.yml allowlists. High-severity drift triggers guided remediation.
Morning briefing (session-briefing.sh)	When a session opens	Missing context — a 12h-throttled digest of what changed across the team since you last looked.
Checkpoint recovery (checkpoint-recover.sh + session-cleanup.sh)	Session start + end	Lost project narrative — a session that ends without a /checkpoint leaves a marker, and the next session offers recovery from git history.
Docflow entry (block-direct-conversion.sh)	Before raw pandoc / unzip / soffice / pdftotext Bash calls	Refuses direct conversion commands that would bypass the audited /docflow pipeline.
Skill integrity watch (skill-creator-watch.py)	Before edits under .claude/skills/	Uncontrolled changes to the shared skill surface — skill edits follow the skill-creator conventions.
Taxonomy freshness (taxonomy-freshness.sh)	When a session opens	Silently stale doctype-governance mappings — flags .taxonomy.yml files past their review cadence.
Usage telemetry (usage-metrics-refresh.sh + usage-metrics-publish.sh)	Session start + end	Unmeasured spend — collects per-session token usage locally and publishes it to the shared metrics branch.

Rules — project conventions the tool reads and obeys

A **rule** is a short written convention stored in `CLAUDE.md` or `.claude/rules/` that the AI assistant reads before acting. Rules are how we encode agreements like “every folder under `docs/` must have a README” or “every README has a `## Conventions` and a `## Changelog` section” — and then the assistant actually follows them, every time, without being reminded.

One task, one file.

All design work, analysis, and drafts produced under a task live inside the task’s numbered markdown file — never as ad-hoc sibling files. Keeps the trail auditable.

READMEs have `## Conventions` and `## Changelog` sections.

When Claude edits a README, it appends a changelog row explaining the rationale — AI sessions collapse many edits into one commit, and the changelog captures context git alone doesn’t.

Capture strategy and lessons in real time.

Strategic decisions go into the active task as tagged strategy blocks in the same turn they were made — not as a deferred cleanup pass. Same rule for lessons.

Never fabricate standard, clinical, or regulatory content.

Anything not derivable from a distilled source under `docs/external/` or `docs/internal/source-md/` is flagged `[VERIFY]` inline.

Read the contract before reasoning about behavior.

Claims about how a skill, schema, or agent behaves must be grounded in its contract file (`SKILL.md`, `schema`, `prompt`) with line citations — never inferred from names or pattern memory.

Regulator-facing documents follow the authoring standard.

Any edit to a filed DHF or submission body runs the regulatory-authoring lint, a substance-preserving copy-edit, and a QA-conformance pass; citation-bearing edits get an independent reference audit.

Rules — project conventions the tool reads and obeys (cont.)

Derived artifacts are never the source of truth.

Audience-facing retellings (articles, decks, knowledge packs) are re-derived from canonical documents, never worked from or "fixed" in place — the controlled source always wins.

Every change lands through a pull-request record.

Even with no human review gate, work merges to main via a PR — a durable, linkable audit trail of what changed, when, and why.

AI provenance without vendor lock.

AI-assisted edits to controlled documents are logged in a non-published metadata block, attributed only as "AI assistant(s)" — the controlled record stays human-authored and tool-agnostic.

Demo banner on every fabricated datum.

Every illustrative clinical figure, placeholder predicate, or sample analysis carries `_Demo sample data – not for clinical use._` near the top.

PRINCIPLE TILES

The regulated-document authoring pipeline

**Lint.**

A deterministic pass catches mechanical defects — internal leaks (task IDs, repo paths), hedge...

**Copy-edit.**

A substance-preserving redline improves clarity and register without touching claims, classifications, numbers, or citations.

**QA-conformance.**

An independent pass verifies the document against its governing QMS form and the three-tier...

**Independent reference audit.**

For citation-bearing edits, a separate agent re-derives every citation from the byte-correct source —...

QUALITY & PROCESS · THE SUM

Every change is, by construction —

- 1 · Owned

By a named, dated task document — the task gate makes ownership automatic.
- 2 · Located

In a folder whose rules were read before the write.
- 3 · Traced

To a design input, architecture node, V&V item, and risk control.
- 4 · Audited

Against a standing best-practices registry.
- 5 · Security-checked

At session start, against the project’s tool allowlists.
- 6 · Recoverable

Every session’s narrative is checkpointed; a dropped session auto-recovers.
- 7 · Measured

Token spend and modeled hours-saved, rolled up per task.

None of the seven depends on a person remembering. **Compliance stops being a tax on velocity and becomes a property of the toolchain.**

06

Humans in Charge

Where the handoff lands. Agents advise; humans decide; the regulated record is human-attributed.

PRINCIPLE TILES

What an "agent" actually is here

**Read only the documents a real person in that role would read**

FDA guidance, ISO/IEC standards, the project's own design controls, the predicate analysis, the relevant...

**Answer within the scope of that role**

and say "that's outside my lane — ask the XX advisor" when a question...

**Cite the sources**

for every claim, so the human reviewer can verify.

**Flag disagreement**

with a draft rather than rubber-stamp it.

PRINCIPLE TILES

How it helps with complex problems



Cybersecurity

(IEC 81001-5-1, signing-trust model, FDA §524B)



Risk management

(what happens if the push fails mid-update? ISO 14971 hazard analysis)



Systems engineering

(where the SaMD / MDDS boundary sits; impact on composition manifest)



Regulatory

(does this change PCCP envelope scope? Does it alter the MDDS classification of the...)



Clinical

(does it introduce dose-limit latency a clinician must account for?)



V&V

(what new test cases does it require? Do they belong in filing-scope or commercial...)

PRINCIPLE TILES

Humans decide. Agents advise.



Reviews the analysis.



Chooses the answer (the agent may have given three).



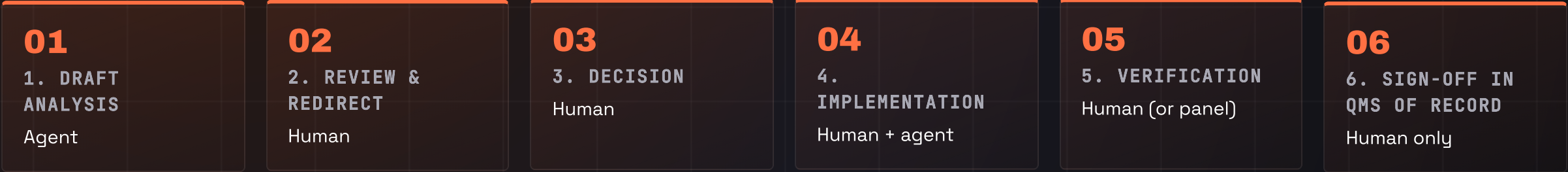
Writes the decision into the task document.



Signs off in the QMS of record when the decision moves to a controlled artifact.

HANDOFF RELAY

Where the handoff happens



The AI never occupies step 3 or step 6. That is not a limitation — that is the design.

PRINCIPLE TILES

Why this works for regulated work



A human-signed decision.



In a task document owned by a named person.



Backed by a trace matrix rebuilt from source.



Supported by a structured analysis with cited sources.



Vetted by multiple specialist perspectives.



Produced inside a toolchain whose compliance properties are enforced by hooks, not hope.

07

The Project Console

The single pane: advisors, documents, task activity, dashboards, trace, strategy review, the FDA package, gap analyses, team metrics, and project settings.

How the console relates to Claude Code

Same agents. Same grounding. Two runtimes. Updating an agent (`.claude/agents/regulatory-affairs.md`) updates both surfaces simultaneously — no drift between what Claude Code sees and what the console exposes.

SURFACE	RUNTIME	GROUNDING	AUTH
Claude Code main thread	CLI, developer workstation	Direct filesystem access	Local
project-console chat	Browser, FastAPI + Claude Agent SDK	Same agent files under <code>.claude/agents/</code> via shared <code>advisors_lib</code> symlink	OAuth to Claude CLI; domain-restricted per <code>project.yml</code> → <code>security.approved_email_domains</code>

01 Landing Page

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Overview

Strategy

Submission

Dashboards

Trace Matrix

Gap Analysis

Workflows

Agents

Documents

PDLC_DEMO

patient-controlled-analgesia-pump · 510k

Overview

Guided tour of the program

Device, regulatory strategy, project shape, and the agentic approach.

Agents

31 domain agents

Chat with individual KOLs or panels, grounded in project documents.

Documents

Browse the project tree

Navigate DHF artifacts, view files, and auto-summarize with Claude.

Dashboards

Submission tracker and readiness views

Live rollups of DHF, submission, and engineering artifacts.

Workflows

Automation and process templates

Coming soon.

A GlobalLogic demo of agentic PDLC workflows in MedTech.

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Privacy

Compliance & Disclosures

One local console over the same agents and artifacts the engineering environment works with.

02 Agents

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Overview

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Agents

Domain agents grouped by function. Chat with individuals or convene a panel.

All

Key Opinion Leaders

Core Team Assistants

Red Team

Key Opinion Leaders

Clinical and technical thought leaders informing PP3500 (PainEase PCA Advanced) design inputs, usability, and validation. Each KOL is grounded in their own profile plus the portfolio device registry and regulatory filings.

PANELS

PP3500 KOL Panel (LLM-moderated)

Panel · 4 members

LLM-moderated advisory panel of PP3500 KOLs — a silent moderator picks the next speaker each turn based on the conversation, instead of rotating round-robin.

PP3500 Primary KOL Panel

Panel · 4 members

Round-robin advisory panel of the four KOLs most relevant to PP3500 — PCA clinical lead, usability, alert fatigue, and infusion nursing.

INDIVIDUALS

Susan Braithwaite

KOL — insulin infusion protocols, ICU glucose management (portfolio context, not PCA-specific).

Kathleen K. Giuliano

KOL — IV smart pump usability, medication administration safety, human factors.

Lisa Gorski

31 grounded advisors — every answer cites the project document it came from.

03 Documents Explorer

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Project tree

Collapse all

docs

_analysis

external

internal

project

console

dhf-manifest

dhfs

input-analysis

competitive-landscape

kol-feedback

market-research

predicate-analysis

README.md2.0 KB

milestones

engineering.yml11.8 KB

README.md2.1 KB

regulatory.yml4.3 KB

strategies

architecture-strategy....13.0 KB

commercial-strategy.md19.7 KB

development-strategy....1.1 KB

operations-strategy.md1.1 KB

postmarket-strategy.md1.2 KB

README.md5.0 KB

strategic-market-ai-infusion.md

docs/project/input-analysis/market-research/strategic-market-ai-infusion.md

MARKDOWN

6.6 KB

.MD

Download

Open in new tab

AI summary

Summarize

Click Summarize to generate a short AI summary of this file.

Strategic Market and Clinical Analysis for AI-Driven Infusion Therapy Portfolio: Competitive Positioning and Regulatory Pathway Optimization

Demo sample data — not for clinical use. Illustrative content for the PDLC_DEMO project (anchor product: PP3500 / PainEase PCA Advanced).

Source PDF: source/Strategic Market and Clinical Analysis for AI-Driven Infusion Therapy Portfolio_ Competitive Positioning and Regulatory Pathway Optimization.pdf

Pages: 26

Converted: 2026-04-12

Relevance to PP3500: Medium — focuses on AI-enabled smart pumps, ambulatory wearables, and neonatal precision; PCA is mentioned only by extension via the broader infusion-pump portfolio.

Executive Summary

This Tier 1 strategic analysis validates the development priorities for an AI-driven infusion therapy portfolio across three high-growth segments: AI-driven predictive monitoring, ambulatory wearable infusion platforms, and neonatal precision technology. The combined Year-5 revenue potential is estimated at ~\$350M, with portfolio-level IRR of 58% and NPV of \$425M against a \$180-240M revenue projection aligned with the AI-enabled medical device market.

The AI-driven predictive monitoring segment is the most transformative opportunity: validated clinical evidence reportedly

Ask the advisor

The whole project tree, rendered and summarized, with an advisor drawer on every file.

04 Tasks

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ACTIVITY SUMMARY

Tasks

What's moving and what's open — derived by `/task summary` from the task docs, not a task list. generated today

7

IN PROGRESS

3

QUEUED

87

COMPLETE

107

ALL TIME

THE LAST 30 DAYS

A tooling-heavy month: the usage-metrics value story went from raw token counts to a red-teamed Value & ROI view in the console — collection, retrospective cost allocation, and the honest modeled-not-measured framing all landed — while the console itself kept growing: the Setup tab arrived and expanded into a full project-settings surface, now joined by this Tasks view. On the regulatory side the reference registry gained the 21 CFR 820 QMSR set and the accessories guidance, and routine registry syncs kept skills in lockstep with hitachi. The open board is small and mostly console/skills work; the one substantive DHF workstream in flight is the PP3500 risk-file backfill with trace-matrix risk mapping (ben/102).

⚠ Watch: ben/046 — the PCA Infusion System filing-entity / multi-function-device architecture gap — remains unstarted while DHF work continues around it; it shapes the filing structure everything else anchors to.

12 closed

16 created

17 touched

since 2026-06-20

OPEN NOW 10

📌

SECOPS Roster Reconciliation

In Progress

ben/055

open 68d

High

🔧

Agent Symlink Fix + sync-skills Windows Guard + Rule

In Progress

Restored 14 corrupted `.claude/agents/*.md` symlinks broken by upstream commit `07574e9` (filemode 120000 with 5KB target string content). G1: corrective PR rewriting the 14 entries via pre-vlad blob...

ben/066

open 51d

High

RECENTLY SHIPPED 12

2026-07-15

📦

Reference Registry Extension: 21 CFR 820 (QMSR) + Missing Standards Distillations

ben/105

All 6 agents returned; 12 standards files + READMEs + audit-report reconciliation landed. Headline: IEC 60601-2-24 non-recognition + public front-matter...

2026-07-14

📦

Risk File Backfill + Trace-Matrix Risk Mapping

ben/102

Contracts read (trace-matrix SKILL.md v8, risk.py default parser, graph.py overlay model, GL-TMP-RM-003/-004 FORMs, GL-STD-RM-001 criteria, 34-DI design-...

🔧

Registry Sync 2026-07-14 + Statusline Install + Submissions VERSION Fix

What's moving and what's open — derived from the task docs, not a separate tracker.

05 Submission Tracker

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Submission Package Tracker

No active session

0 pending

Pending changes

Save & Publish

Cancel session

Source

Last updated: 12-May-26

QSUB

38

0 approved · 13 in flight · 17 not started · 0% ready

510K+PCCP

44

0 approved · 13 in flight · 23 not started · 0% ready

LMR1

33

0 approved · 13 in flight · 12 not started · 0% ready

LMR2

39

0 approved · 13 in flight · 18 not started · 0% ready

ENGINEERING PREREQS

20

0 approved · 0 in flight

GRAND TOTAL

154

0/154 approved · 0%

Approved

In Review

Drafted

Drafting

Needs Revision

Not Started

Click row for help

QSub

13

0/38 done

510k+PCCP

13

0/44 done

LMR1

13

0/33 done

LMR2

13

0/39 done

Expand All

Collapse All

Show All Info

Show All Help

STATUS

All

SCOPE

All

PHASE

All

EFFORT

All

QSUB

(all bindings) (5)

5

ID	DELIVERABLE	SCOPE	PHASE	REF	EFFORT	AI STATUS	STATUS	LAST UPDATED	PATH
Q4	Q-Sub cover letter	Suite	QSub	FDA Q-Sub Guidance Appendix 1	Low		Not Started	—	Create Draft
Q5	Device description (Q-Sub formal)	Suite	QSub	FDA Q-Sub Guidance § Content requirements	Med		Not Started	—	Create Draft
Q6	Proposed indications for use	Suite	QSub	FDA Q-Sub Guidance § Content requirements	Low		Not Started	—	Create Draft
Q7	Predicate device analysis (Q-Sub summary)	Suite	QSub	FDA SE Guidance (2014) §III	Med		Not Started	—	Create Draft

Ask the advisor

154 deliverables across Q-Sub and 510(k)+PCCP scopes – status editable in the browser.

06 Trace Matrix

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← All DHFs

Trace Matrix — pca-device

Assembled 2026-07-15T06:55:42Z ?

Rebuild

User Needs 22

Design Inputs 34 9 1

SW Reqs 32 9 28

Architecture 7 9 7 ~

VnV 3

Risk 16

CRITICALITY

All

CtS

CtF

CtC

CtP

Supp

Untagged

All rows

Orphans only

Expand all groups

Collapse all groups

ID	REQUIREMENT	STATUS	CRIT	← UN	→ SW	→ V&V	↔ RISK ⓘ
G1 — Therapy Delivery 4 items							
DI-001	The device shall deliver programmed continuous flow rates from 0.01 to 25 mL/hr with an...	—	CtS	UN-001 UN-002	SW-001 SW-005	VER-PP3500-BT-005	HAZ-001
DI-002	The device shall deliver patient-activated bolus doses from 0.01 to 20 mL with a...	—	CtS	UN-001	SW-002 SW-005	—	—
DI-003	The device shall support a programmable lockout interval from 1 to 99 minutes between...	—	CtF	UN-003	SW-003	VER-PP3500-SW-002	HAZ-014
DI-004	The device shall enforce programmable 1-hour and 4-hour cumulative dose limits and shall...	—	CtS	UN-003 UN-004	SW-004	VER-PP3500-SW-002	HAZ-001 HAZ-002 HAZ-014
G2 — Drug Library & Medication Safety 3 items							
DI-005	The device shall enforce drug-library hard maximum limits such that no programming path —...	—	CtS	UN-004 UN-009	SW-005 SW-006	VER-PP3500-SW-006	HAZ-002 HAZ-003
DI-015	The device shall maintain an on-board drug library of at least 200 medications with...	—	CtF	UN-009	SW-007	—	HAZ-003
DI-016	The device shall include an integrated 1D/2D barcode scanner capable of reading...	—	CtF	UN-008	SW-008	—	HAZ-003
G3 — Alarms & Annunciation 4 items							
DI-007	The device shall detect downstream occlusions in the range 3 to 15 psi and annunciate a...	—	CtS	UN-006	SW-009	—	HAZ-006
DI-008	The device shall detect single air-in-line bubbles ≥50 µL and cumulative air ≥1 mL over a...	—	CtS	UN-006	SW-010	—	HAZ-007
DI-009	The device shall annunciate a high-priority over-infusion alarm and	—	CtS	UN-001	SW-011	—	HAZ-001

Ask the advisor

All ten DHFs traced across six layers, from user needs to risk.

07 Strategy Review

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Strategy

Review + promote strategy decisions per domain — Accept / Reject / Modify / Re-Assemble all live.

Live.

Pick a strategy on the left, review pending proposals, and take action. Every Accept / Reject / Modify / Re-categorize updates the strategy and adds a History note. Re-categorize also saves and publishes automatically. We need to know who's making the change, so make sure your project task is active.

DOMAIN	STATUS	LAST ASSEMBLED	PROPOSALS	HISTORY
Architecture Strategy	assembled	2026-04-14	0	1
Commercial Strategy	assembled	2026-06-03	0	1
Development Strategy	awaiting-content	—	0	0
Operations & Tooling Strategy	awaiting-content	—	0	0
Post-Market Strategy	awaiting-content	—	0	0
Regulatory Strategy	assembled	2026-04-14	0	1
Risk Strategy	awaiting-content	—	0	0
Testing & Validation Strategy	awaiting-content	—	0	0

docs/project/strategies/architecture-strategy.md · status assembled · last assembled 2026-04-14 · sources: task 006, task 009

0 proposals — nothing to review

RUN ASSEMBLER

Scans your task docs for new STRATEGY CONTENT blocks, merges them in, and surfaces conflicts here as proposals.

Run Assembler →

Takes a few minutes; uses AI credits. The result lands as a diff you can review, then Save & Publish or Cancel to revert.

Ask the advisor

Eight strategy domains as a live review surface — accept, reject, or modify harvested proposals.

08 Submission Package

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FDA SUBMISSION PACKAGE

Refresh

Submission

Pre-Submission (Q-Sub), 510(k), and PCCP packages – composition manifest, FDA-facing documents, and the questions for FDA. Each filing is rendered from the submissions skill's sidecars; open one to read the package and ask the regulatory advisor about it.

2
FILINGS

7
DOCUMENTS

1
BLOCKING

Scaffold510(k)

510(k) – PP3500 (PainEase PCA Advanced)

docs/project/submissions/510k

0
DOCS

26
REQUIRED

0
BRIEFS

0
QUESTIONS

Open package →

DraftingQ-Sub (Pre-Submission)⚠ 1 blocking

Q-Sub (Pre-Submission) – PP3500 (PainEase PCA Advanced)

docs/project/submissions/qsub

7
DOCS

7
REQUIRED

2
BRIEFS

6
QUESTIONS

Open package →

A GlobalLogic demo of agentic PDLC workflows in MedTech.

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Privacy

Compliance & Disclosures

The FDA-facing package: composition manifests, documents, and open questions in one place.

The project critiques its own work product against standards, with advisor panels per finding.

10 Value & ROI

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Team Metrics

Claude Code token usage & estimated cost across the team – aggregated and **anonymized** (Member N). Built from `tools/usage-metrics/usage.json`. Token quantities in **millions (MTok)** to match the \$/MTok rates.

Token Metrics

Value & ROI

Hours saved = the by-hand person-hour *estimate* (ranged) minus **how long it actually took us** (the agentic time). **Category** is the by-hand specialist persona(s). **Agentic cost** is the *measured* token spend (real rates). Headlines lead with the conservative low end; people anonymized.

Range

All

Last 30 days

Last 60 days

Last 90 days

PERSON-HOURS SAVED

923.2–3,052.2 hrs · 65–86% of by-hand

BY-HAND EFFORT

1,429–3,542 hrs

AGENTIC TIME

489.8–505.8 hrs (est.)

AGENTIC COST

\$623 tokens

TASKS VALUED

104

BY CATEGORY — WHERE THE SAVINGS COME FROM

All Categories · 104 tasks

Hours saved923–3,05265–86%

By-hand effort1,429–3,542 hrs

Agentic time490–506 hrs

Agentic cost\$623

Rd-Lead · 83 tasks · 80% of tasks

Hours saved460–1,64659–84%

By-hand effort785–1,961 hrs

Agentic time314–324 hrs

Agentic cost\$358

Quality-Engineering · 12 tasks · 12% of tasks

Hours saved127–36678–92%

By-hand effort162–399 hrs

Agentic time34–36 hrs

Agentic cost\$51

Regulatory-Affairs · 17 tasks · 16% of tasks

Hours saved114–35371–88%

By-hand effort160–399 hrs

Agentic time46 hrs

Program-Manager · 17 tasks · 16% of tasks

Hours saved73–24862–85%

By-hand effort118–293 hrs

Agentic time44–46 hrs

Systems-Engineering · 10 tasks · 10% of tasks

Hours saved59–18368–87%

By-hand effort86–210 hrs

Agentic time26–27 hrs

104 tasks · 920–3,050 modeled person-hours saved vs ~\$623 token spend – modeled, uncalibrated.

11 Project Settings

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PROJECT SETTINGS

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Environment

Read from `project.yml`,
`.mcp.json`, and `.claude/`.
Editable here: connectors, registry
installs, team roster. Skills and
agents are managed by the registry
sync tooling; machine changes stay
in your terminal.

Connectors

MCP servers available to Claude Code in this project — merged from `.mcp.json` (definitions) and `project.yml` `security.approved_mcps` (the team's allowlist). Environment variable values are never shown or edited here.

Connector changes take effect at the next Claude Code session start (or via `/mcp` → Reconnect) — the console edits configuration only; it never launches or stops MCP servers.

POPULAR

Chrome DevTools

configured

Browser automation and inspection (navigate, screenshot, console, network) via the chrome-devtools MCP server.

Reference

View configured ↓

File Locator (semantic search)

configured

Local semantic file search over the project's docs corpus (fastembed + SQLite FTS5). Ships with the file-locator skill; requires the skill to be installed and its index built.

Reference

View configured ↓

Atlassian (Jira / Confluence)

Atlassian's official remote MCP server — Jira, Confluence, and Compass tools over OAuth. Sign-in happens in the MCP client on first connect (`/mcp` in a session); no tokens in config.

Reference

Configure...

Custom stdio server

Any MCP server launched as a local process (command + args). Use for servers not in the catalog.

Reference

Configure...

Custom HTTP / SSE server

Any remote MCP server reachable over HTTP (streamable HTTP or SSE). Authentication is handled by the MCP client, not the console.

Reference

Configure...

CONFIGURED

CONNECTOR	TYPE	STATUS
chrome-devtools	stdio	✓ ▶
file-locator	stdio	✓ ▶

Every skill, agent, hook, and registry audited against the project manifest.

12 Workflows

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Workflows

Composed automation across project skills. Each workflow declares its backend readiness — look-and-feel prototypes are clearly labeled.

Readiness: **live** (wired end-to-end) **prototype** (some placeholder zones, clearly labeled) **proposed** (UI-only, backend not wired)

Gate-Readiness

A1 DHF Gate Check

Ready/not-ready verdict for a single DHF — composes manifest gaps, trace validation, and BP audit.

preview — not yet clickable-through

A2 Submission Pre-Flight

Per-filing go/no-go — composition manifest + tracker + docflow round-trip + secops attest.

preview — not yet clickable-through

A3 PR Readiness (pre-ultrareview)

Blocking + non-blocking issue list before kicking off /ultrareview.

preview — not yet clickable-through

Authoring

B1 Doc Round-Trip Batch

Pick N markdown files → preview the /docflow export plan → (later) execute.

B2 Trace Refresh on Save

Edit a requirements / UN / V&V file → one-click scoped /trace-matrix rebuild.

preview — not yet clickable-through

B4 Tracker Status Update

Click a Status badge in the submission-tracker dashboard → pick a new value → accumulate pending changes → Save & Commit writes back to submission-tracker.md, regenerates HTML, commits + pushes, closes the

Composed automation with honest readiness labels: live, prototype, or proposed.

— QUESTIONS & NEXT STEPS —

Thank you.

Let's bring a **purpose-built agentic harness** to your MedTech programs.

Ben Xavier • CTO HCLS GlobalLogic • 2026

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